

DRAWING LIST FOR TEST PACKAGE**WA-FR1-631****PROJECT NO.: 22328D****PROJECT TITLE : M15X Ph2 Project**

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	631	R2FA	150A-WA-FR1-N-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
2	631	R2FA	200A-WA-FR1-N-7008-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
3	631	R2FA	200A-WA-FR1-N-7009-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
4	631	R2FA	150A-WA-FR1-N-7010-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
5	631	R2FA	150A-WA-FR1-N-7011-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
6	631	R2FA	150A-WA-FR1-N-7012-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
7	631	R2FA	150A-WA-FR1-N-7013-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
8	631	R2FA	150A-WA-FR1-N-7014-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
9	631	R2FA	150A-WA-FR1-N-7015-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
10	631	R2FA	200A-WA-FR1-N-7016-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
11	631	R2FA	150A-WA-FR1-N-7017-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
12	631	R2FA	150A-WA-FR1-N-7018-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 506	2A/1A	
13	631	R2FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
14	631	2FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	
15	631	3FA	400A-WA-FR1-N-7019-01	1 OF 1	0	CJ-M15W14-PR-A20-000168 / 505	2A/1A	



INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ²)	TEMP. (°C)	AREA	W14						
	THICK.	—	DESIGN	6.4	60	UNIT	R2FA						
			OPERATING	4	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-020015-A		25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동호	
						REF. P & ID							
						CJ-M15W14-PR-A20-000505		REV. DATE	DESCRIPTION	BY	CHK	APP'T	

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY WA		JOB NO. 22330D	
LINE NO.	150A-WA-FR1-N-7007	SH'T NO. 01 / 01	REV. 0



* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COLLARED: HC	* ECC RED: RE	* CON SWAGE: RCS/RCS1	* ECC SWAGE: RES/RES1	* NIPPLE: NP/NP1/MP2	* CAP: CAP/CAP1	* FLUG: FLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET (HPIPE/PIPE): SC	* SADDLE (HPIPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VAL: VG/VG1/VG2	* BALL VAL: VB/VB1/VB2/VB3/VB4	* BUTTERFLY VAL: VF/VF1	* CHECK VAL: VC/VC1/VC2/VCS/VCS1/VCS2/VCS3	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH		
* GLOBE VAL: V/G1/V1/2	* DIAPHRAGM VAL: VD/VD1	* DAMPER: DP	* QUICK COUPLINGS: SC	* FLEXIBLE JOINT/STEEL: FLXJ	* FLEXIBLE JOINT/RUBBER: FLXJR	* ISOLATION JOINT: IJ	* FLEXIBLE JOINT/STEEL PIPE LINED: FLXJL	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT & NUT: BL/N1/1			
* STRAINER (T-YPE): TSTR	* STRAKE RTD (T-YPE): TSTR	* CHIMNEY (DUCT) TYPE: CSTR	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* EXHAUST KITT (KUBBER): K	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTD: LOTD	* SIGHT GLASS: SG						

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA						
			OPERATING	4	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-020014-A							
						REF. P & ID							
						CJ-M15W14-PR-A20-000505							
								25.01.24	APPROVED FOR CONSTRUCTION	황래림	안남호	박동	
								DATE	DESCRIPTION	BY	CHK	APP	



②

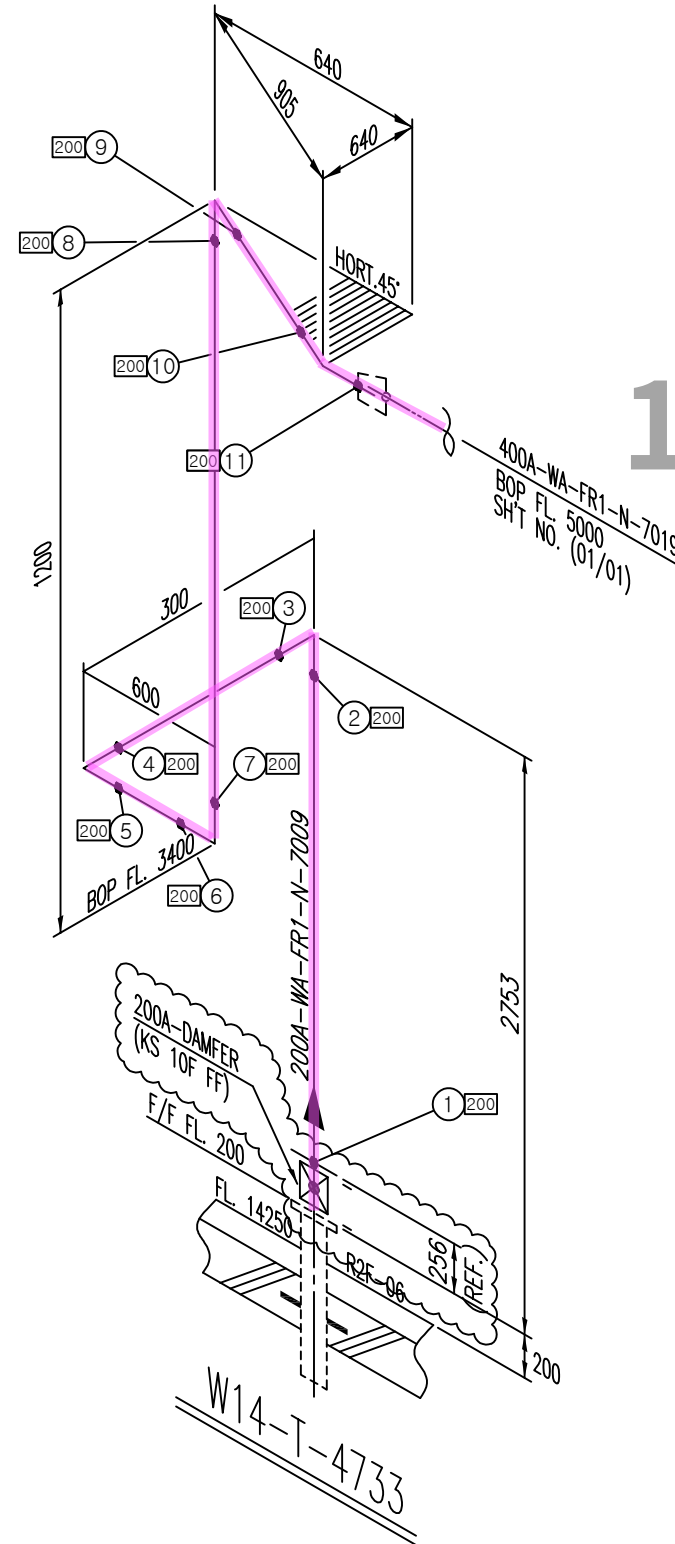
CTOR
SK ecoplant

WA

200A-WA-FR1-N-7008

PROJECT NAME	
M15X Ph4 Project	
JOB NO.	
22330D	
SH'T NO.	REV.
01 / 01	0

경고: 본 도록은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

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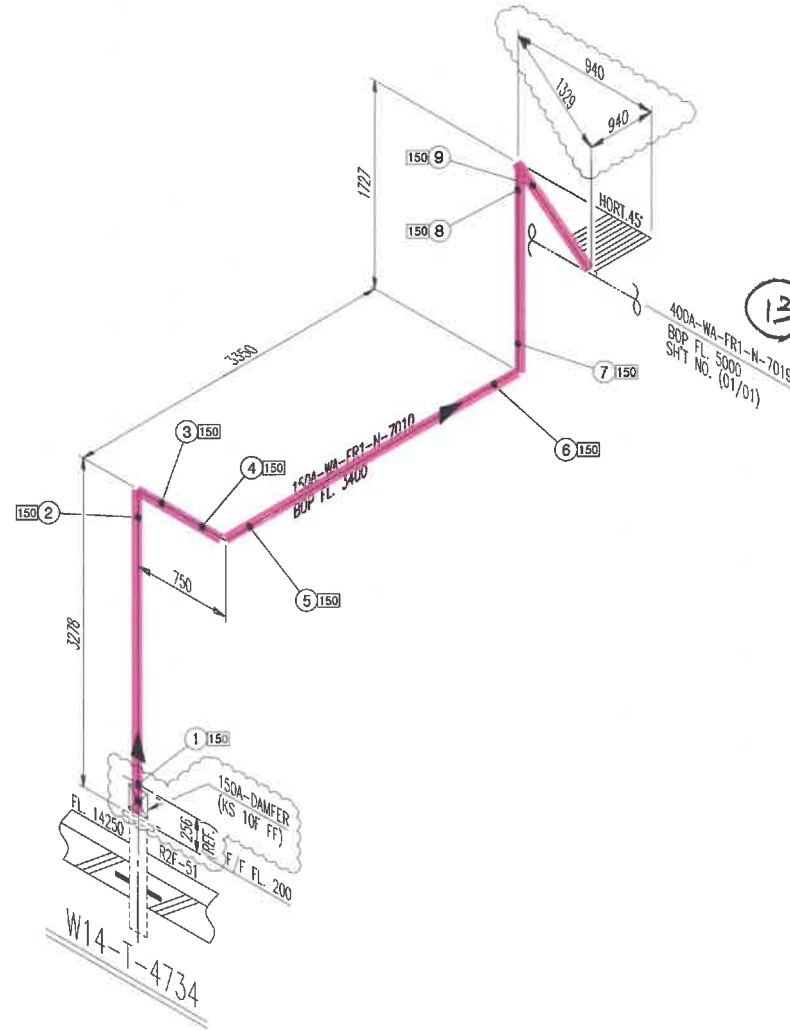
<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB10/VB2/VB210 * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD10 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.4	60	UNIT	R2FA							M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-020014-A							WA	22330D		
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000505		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	200A-WA-FR1-N-7009	01 / 01	0

정고 : 부수면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. TRAY & ANGLE CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

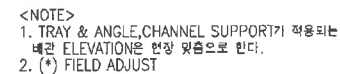
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLTD : RC * ECC RED : RE * CON SWAGE : RCS/RCS1 * ECC SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HDPE/PVC) : SC * SADDLE (HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/G1/G2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DAMP-RAGW VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT (STEEL) : FLX9 * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (Y-TYPE) : YSTR * STRAINER (T-TYPE) : TSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BRID SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	W14	REF. PLAN DWG.	REF. P & ID	REV.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60			R2FA		CJ-M15W14-PI-A30-020014-A	CJ-M15W14-PR-A20-000505		25.01.24	APPROVED FOR CONSTRUCTION	황래홍	안남후	박동영
			OPERATING	4	AMB												



ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	8.4M	PPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	150A	4EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	150	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
G	150A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	150A	2SET	MACHINE BOLT & NUTS :SIS 304 / W WASHERS

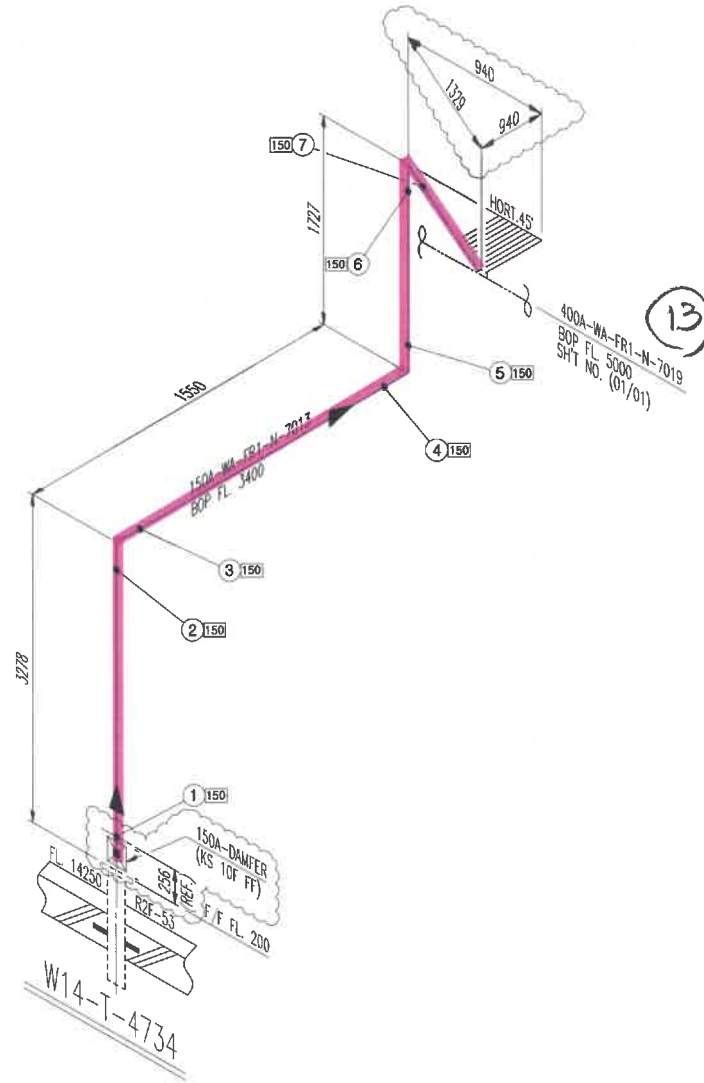
CLIENT	SK hynix	4
CONTRACTOR	SK ecoplant	PROJECT NAME
COMMODITY	WA	M15X Ph4 Project
LINE NO.	150A-WA-FR1-N-7010	JOB NO.
		22330D
		SH'T NO.
		01 / 01
		REV.
		0



* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* COMBED: RC	* ECC.RED: RE	* CON.SNAGE: RCS/RS1	* ECC.SNAGE: RES/RS1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUG: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET/NIPPLE: NSC	* SADDLE/NIPPLE: SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BRID FLANGE: FB/B1	* GATE VA: VG/G1/G2	* BALL VA: VG/BG/AB1/AB10/AB2/AB20	* BUTTERFLY VA: VB/VF/4	* CHECK VA: VC/GCA/MS/ACA/MS/ACA/MS/MS/2					
* GLOBE VA: VG/V1/V2	* DIAPHRAGM VA: VD/VD0	* DAMPER: VP	* QUICK COUPLING: QC	* FLEXIBLE JOINT(STYLE): FLXB	* FLEXIBLE JOINT(RUBBER): FLXR	* FLEXIBLE JOINT(STEEL PIPE LINED): FLXP	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHUTTER: SSH					
* STRAIN NUT: YSTR	* STRAIN NUT-TYPE1: YSTR1	* STRAIN NUT-TYPE2: YSTR2	* ANCHOR PLATE: APL	* FLOOR SCREED: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHELLED: SH	* SPECIAL BOLT & NUT: BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	6.4	60	UNIT R2FA							M15X Ph4 Project		
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020014-A						COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000505	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
								25.01.24	APPROVED FOR CONSTRUCTION	황재중	안남후	박동영	150A-WA-FR1-N-7011	01 / 01	0

참고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

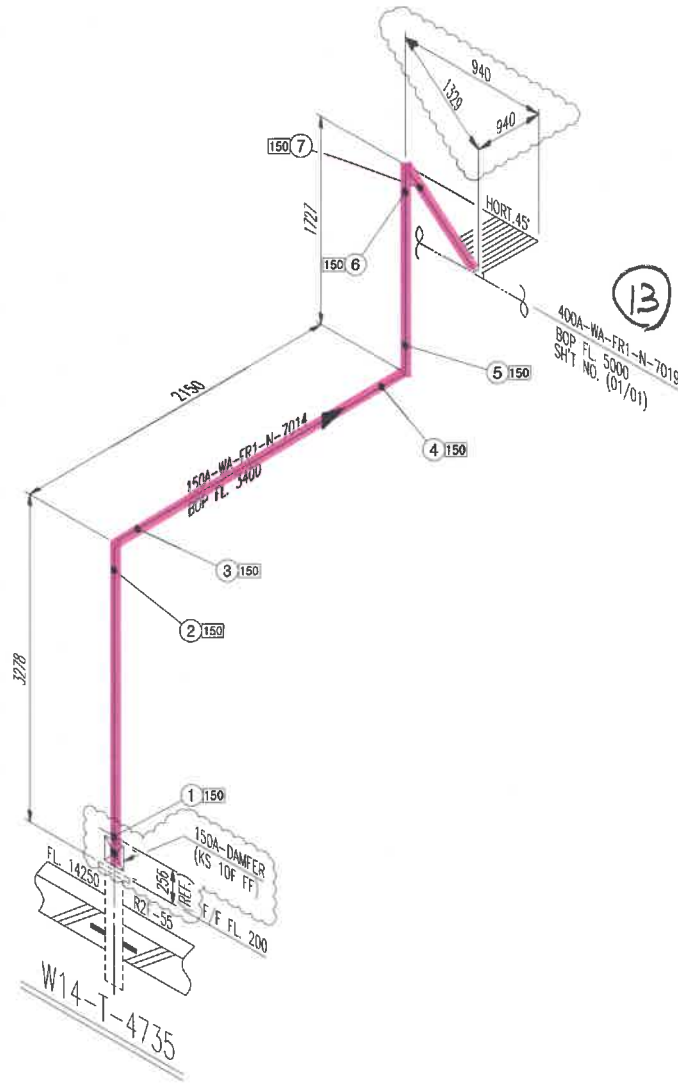
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON. RED : RC * ECC. RED : RE * CON. SWAGE : RCS/HCS1 * ECC. SWAGE : RES/HES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET (HOPE / PVC) : SC * SADDLE (HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT (STEEL) : FLXB * FLEXIBLE JOINT (RUBBER) : FLXR * FLEXIBLE JOINT (STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER (Y-TYPE) : YSTR * STRAINER (T-TYPE) : TSTR * STRAINER (BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BRD. SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ²)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020014-A	REF. P & ID	CJ-M15W14-PR-A20-000505	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60										25.01.24	APPROVED FOR CONSTRUCTION	황래준	안남우	박동영
			OPERATING	4	AMB														

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	6.3M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8T
E90	150A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	150	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
G	150A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM THK
B	150A	2SET	MACHINE BOLT & NUTS :SIS 304 / W WASHERS

CLIENT	SK hynix	7
CONTRACTOR	SK ecoplant	
COMMODITY	WA	
LINE NO.	150A-WA-FR1-N-7013	
PROJECT NAME	M15X Ph4 Project	
JOB NO.	22330D	
SH'T NO.	01 / 01	REV. 0



- <NOTE>
 1. TRAY & ANGLE CHANNEL SUPPORT기 적용되는
 배관 ELEVATION은 현장 모습으로 한다.
 2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLUED : RC * ECC RED : RE * CONSWARE : RCS/RCS1 * ECC SWARE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * STUB-IN : STU * SOCKET(HDPE/PPH) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHUTTER : SSH
 * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020014-A	REF. P & ID	CJ-M15W14-PR-A20-000505	REV.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60										25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동영
			OPERATING	4	AMB														

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	150A	6.9M	PIPE, FRP(MANUFACTURER STD), PE/BUT-JOINT, 4.8T
E90	150A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	150A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	150	1EA	DAMPER, KS 10K, FF, BODY-FRP / ORING-EPDM, LEVER
G	150A	2EA	GASKET, KS 10K, FF, EPDM, 3.DMM TH'K
B	150A	2SET	MACHINE BOLT & NUTS :SIS 304 / W WASHERS

CLIENT

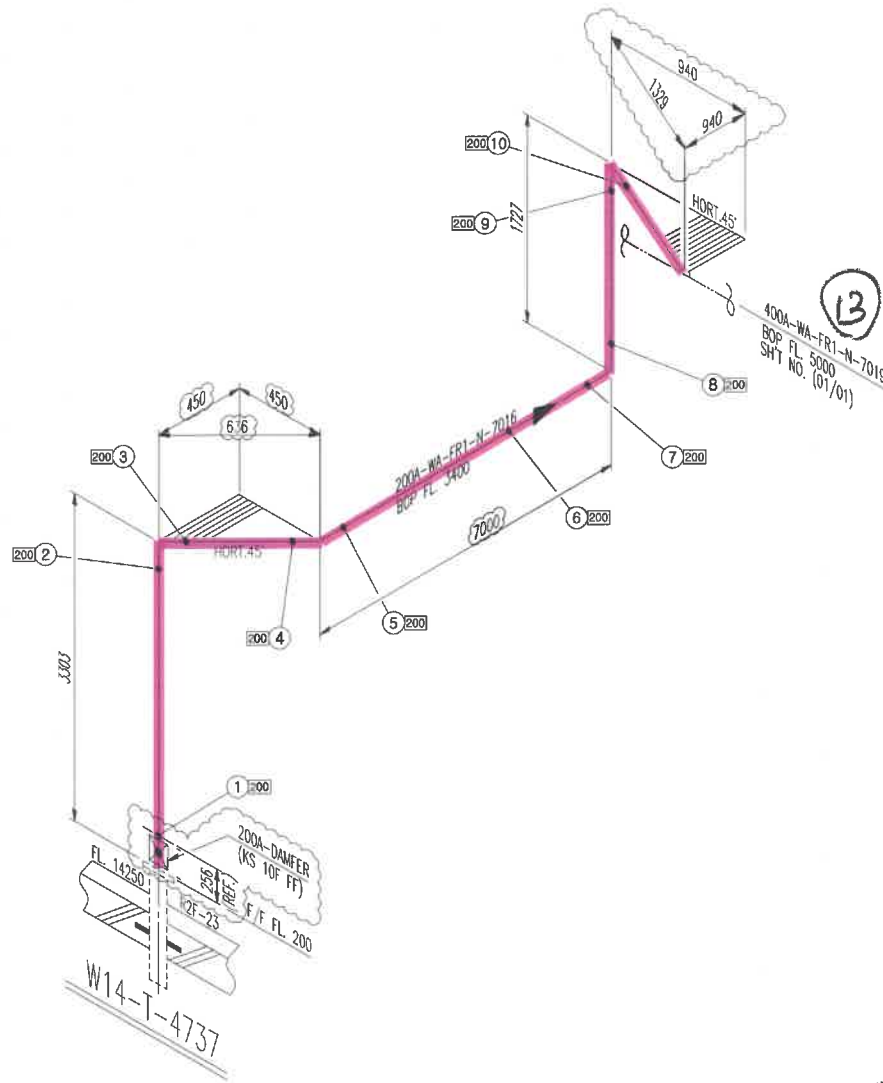
SK hynix

CONTRACTOR	SK ecoplant	PROJECT NAME	M15X Ph4 Project
COMMODITY	WA	JOB NO.	22330D
LINE NO.	150A-WA-FR1-N-7014	SH'T NO.	01 / 01
		REV.	0



0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
REV.	DATE	DESCRIPTION	BY	CHK	APP

CLIENT		 	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.		SH'T NO.	REV.
150A-WA-FR1-N-7015		01 / 01	0



<NOTE>
1. TRAY & ANGLE CHANNEL SUPPORT기 적용되는
내면 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * SUB-IN : STU * REINFORCING PAD : RP * CONVED : RC * ECC RED : RE * CON SWAGE : RCS/W/S1 * ECC SWAGE : RCS/H/S1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* SUB-IN : STU * SOCKET(HOPE/PVC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1/VD2 * DAMPER : VP * QUICK COUPLING : SQ * FLEXIBLE JOINT(STEEL) : FLX * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(I-TYPE) : YSTR * STRAINER(L-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BRID. SCRTEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020015-A	REF. P & ID	CJ-M15W14-PR-A20-000505,000506	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60									25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동영
			OPERATING	4	AMB													

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	200A	11.9M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 4.8I
E90	200A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8I
E45	200A	1EA	ELBOW 45 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2I
F	200A	1EA	FLG, KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / CRWG-EPDM, LEVER
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS-STS 304 / W WASHERS

CLIENT

SK hynix

CONTRACTOR

SK ecoplant

COMMODITY

WA

LINE NO.

200A-WA-FR1-N-7016

PROJECT NAME

M15X Ph4 Project

JOB NO.

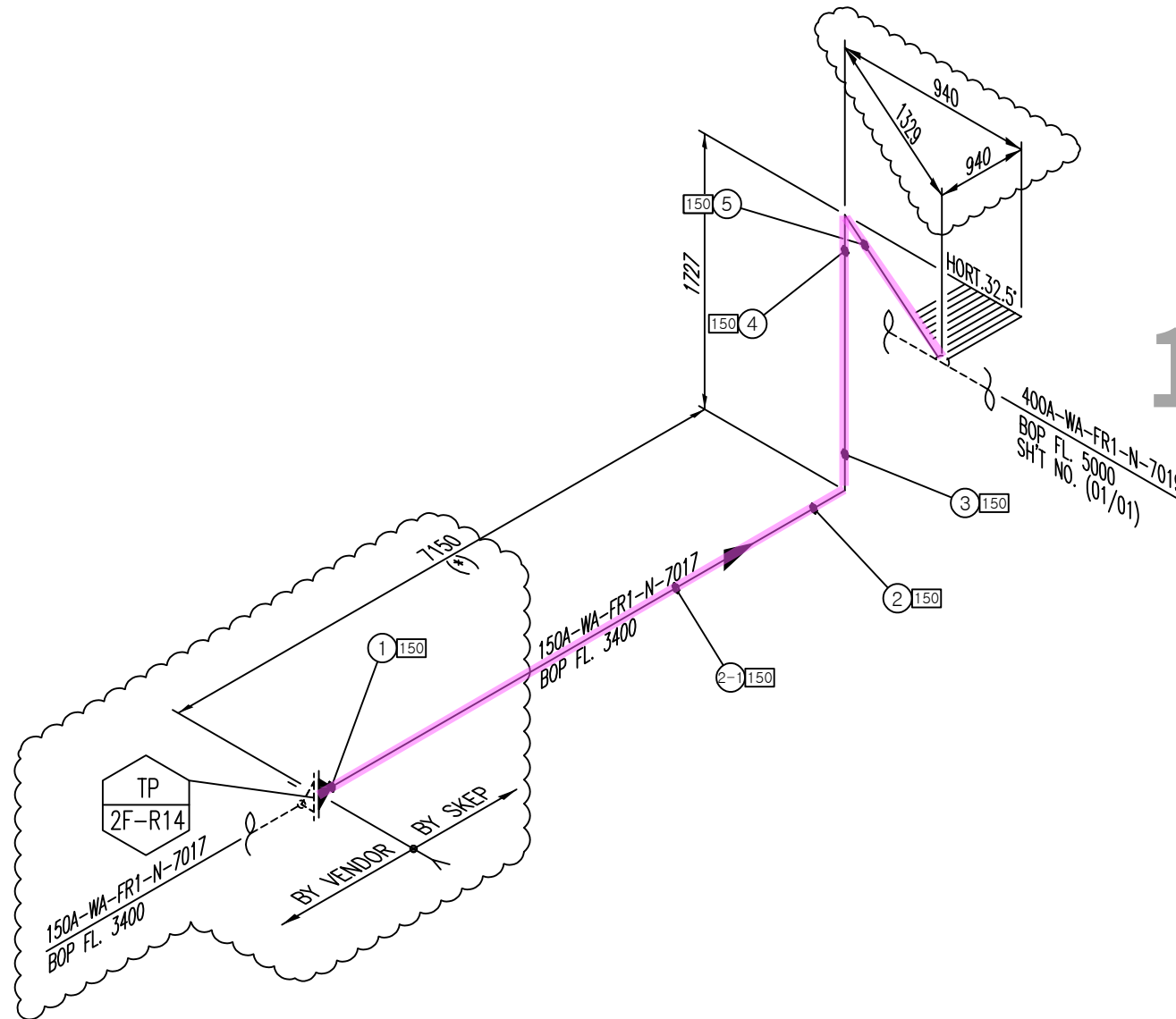
22330D

SH'T NO.

01 / 01

REV.

0

[illegible]

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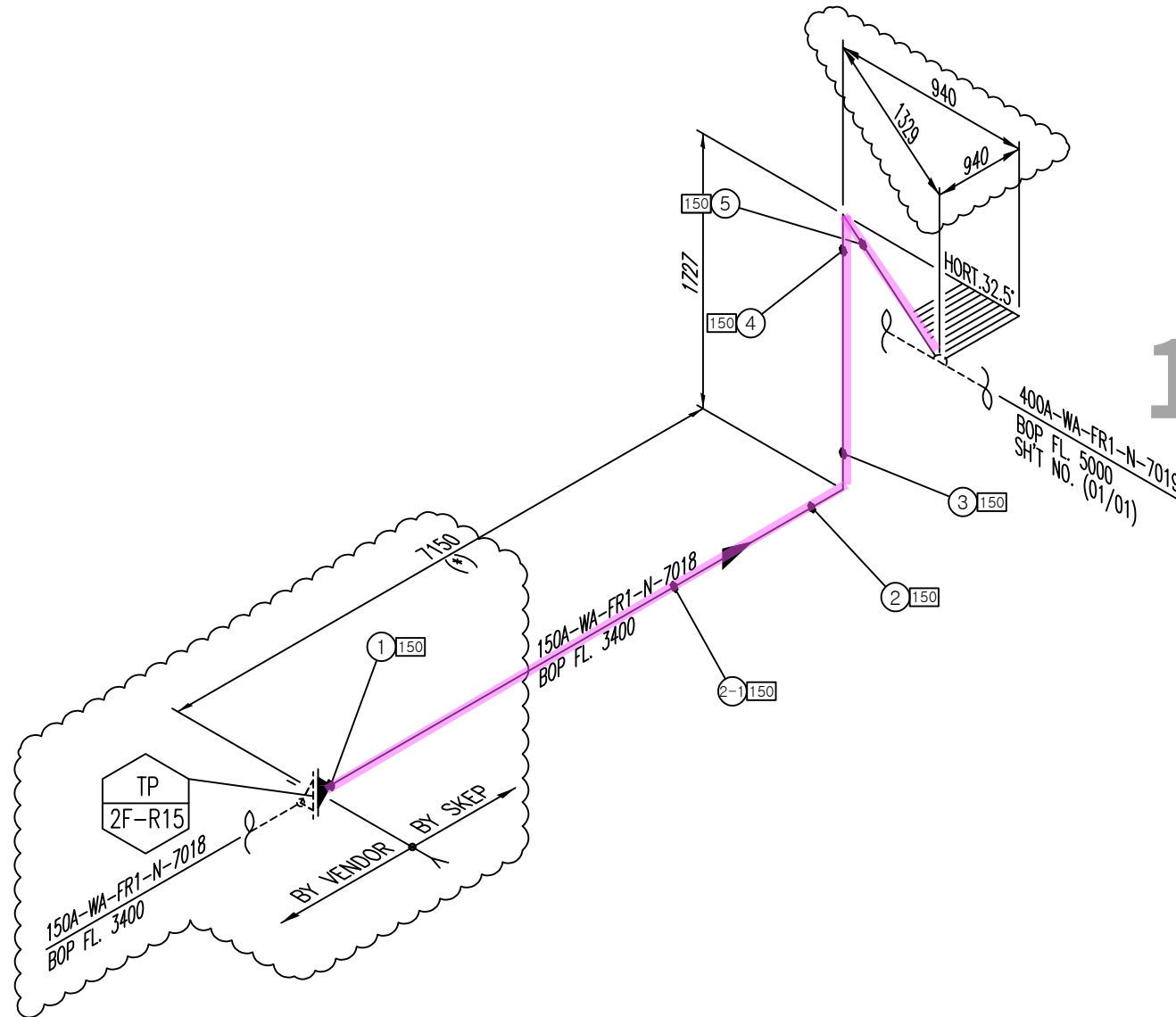
<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.4	60	UNIT	R2FA								M15X Ph4 Project	
			OPERATING	4	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-020014-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	WA	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000505,000506								150A-WA-FR1-N-7017	01 / 01	0

정고: 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840

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<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

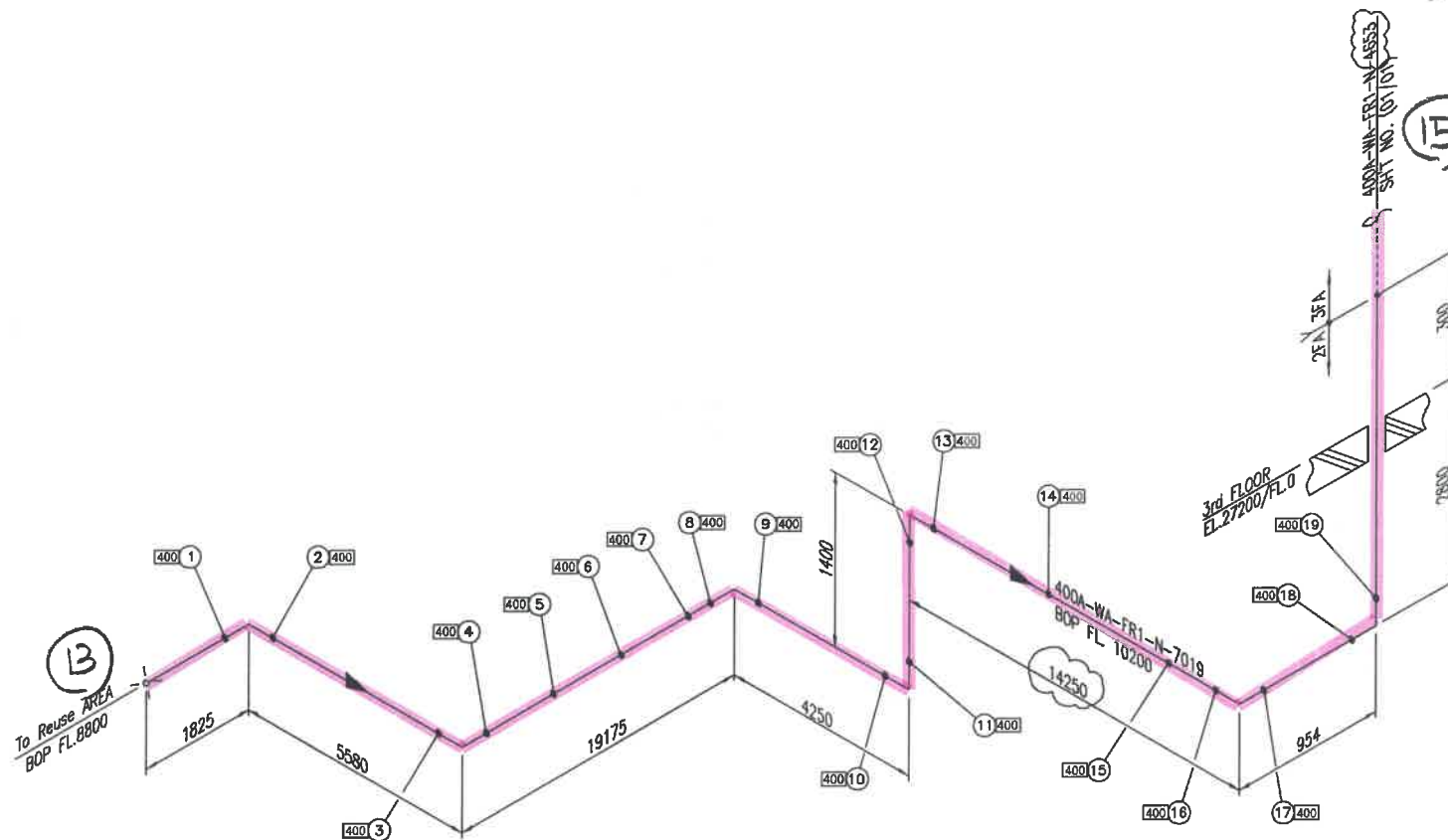
CLIENT



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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.4	60	UNIT	R2FA								M15X Ph4 Project	
			OPERATING	4	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-020014-A								WA	22330D	
						REF. P & ID								LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000505,000506		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		150A-WA-FR1-N-7018	01 / 01	0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

정경호, 최필진, 이원익, 김민준, SK HYNIX의 기밀비밀 자료를로서, 어떠한 경우에도 독점적 외 의 사용 목적 제3자에게 공개, 누설 및 출하할 수 없으며, 회사 내 유·무효포자 불발판 사영으로 피해가 발생할 경우 유·무효포자에게 민·형사상 책임요구를 할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLD : RC * EDCRED : RE * CONSWAGE : RCS/RCS1 * EDCSWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HOPE/PVC) : SC * SHOULDER(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : TB/TR1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB3/VB4 * BUTTERFLY VA. : VV/VV1 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STAP * BRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/B1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14					CONTRACTOR		PROJECT NAME	
	THICK.	-	DESIGN	1	AMB	UNIT	2FA					SK ecoplant		M15X PH4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.						COMMODITY		JOB NO.	
						CJ-M15W14-PI-A30-020011-G						LINE NO.		223300D	
						REF. P & ID						DATE		SH'T NO.	
						CJ-M15W14-PR-A20-000505						DESCRIPTION		01 / 01	
												BY		REV.	
												CHK		0	
												APP'D			

CLIENT

SK hynix

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CONTRACTOR
SK ecoplantCOMMODITY
WALINE NO.
400A-WA-FR1-N-7019PROJECT NAME
M15X PH4 ProjectJOB NO.
223300DSH'T NO.
01 / 01REV.
0

PUNCH LIST

Project No : 22330D
Project Title : M15X Ph3 Project

Package No :
Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning